

SAS Project Submission

Category-3

Q1) What are the types of Data Transformation Phase? Explain the detailing on Data Cleaning/ Smoothing/ Standardization? With each one example on creating?

Ans1) Data Transformation Phase in SAS is a sequence of process of preparing raw data into a clean, structured, and more useable format. Important step for data preprocessing. Data Transformation should be done first before entering into an actual project. **Phases in this step include:** Data Smoothing, Data Concatenation, Data Filtering, Sorting, Splitting Data, Data Cleaning, Data Normalization, Combining Data, Data Aggregation, Standardization, Data Deduplication.

 **Data Cleaning:** Data cleaning identifies incorrect data, handling missing values or incorrect formats and modifies it according to requirements. Can be done using the point and click function: **Tasks and Utilities > Tasks > Data > Sort data/ Filter data/ describe missing values/ Transform data**, or programmatically using the DATA step, PROC SQL, or PROC SORT.

Below is an example on sorting data, here I sorted MSRP of cars in Ascending order, using this function I have sorted the MSRP of cars from lowest prize to highest prize.

SAS® Studio

Server Files and Folders

Tasks and Utilities

- My Tasks
- Tasks
 - Data
 - List Table Attributes
 - Characterize Data
 - Describe Missing Data
 - List Data
 - Transpose Data
 - Stack/Split Columns
 - Filter Data
 - Select Random Sample
 - Partition Data
 - Sort Data
 - Rank Data
 - Transform Data
 - Standardize Data
 - Recode Values
 - Recode Ranges
 - Combine Tables
- Snippets
- Libraries
- File Shortcuts

*Program 1 x *Sort Data x

CODE LOG RESULTS OUTPUT DATA

```
1 data sort;  
2 set sasHELP.cars(obs=20);  
3 run;
```

Line 2, Column 24 UTF-8

SAS® Studio

Server Files and Folders

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- File Shortcuts

*Program 1 x *Sort Data x

Settings Code/Results Split Log Code

CODE LOG RESULTS OUTPUT DATA

```
1 /*  
2 *  
3 * Task code generated by SAS Studio 3.8  
4 *  
5 * Generated on '14/05/2025 07:54'  
6 * Generated by 'u64183310'  
7 * Generated on server 'ODAWS02-APSE1-2.ODA.SAS.COM'  
8 * Generated on SAS platform 'Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64'  
9 * Generated on SAS version '9.04.01N7P08062020'  
10 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/  
11 * Generated on web client 'https://odamid-apse1-2.oda.sas.com/SASStudio/main?locale=en_GB&zone=GMT%252B05%253A30'  
12 */  
13  
14  
15 proc sort data=WORK.SORT out=work.sortDS  
16 (drop=DriveTrain Invoice MPG_City MPG_Highway Weight Wheelbase Length) equals;  
17 by MSRP;  
18 run;
```

Messages: 2 User: u64183310 UTF-8

SAS® Studio

Server Files and Folders

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 - Recode Ranges
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- Snippets
- Libraries
- File Shortcuts

*Program 1 x *Sort Data x

Settings Code/Results Split Log Code

CODE LOG RESULTS OUTPUT DATA

Errors, Warnings, Notes

Errors

Warnings

Notes (3)

```
70 *  
71 * Task code generated by SAS Studio 3.8  
72 *  
73 * Generated on '14/05/2025 07:54'  
74 * Generated by 'u64183310'  
75 * Generated on server 'ODAWS02-APSE1-2.ODA.SAS.COM'  
76 * Generated on SAS platform 'Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64'  
77 * Generated on SAS version '9.04.01N7P08062020'  
78 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/136.0.0.0 Safari/537.36'  
79 * Generated on web client  
79 ! 'https://odamid-apse1-2.oda.sas.com/SASStudio/main?locale=en_GB&zone=GMT%252B05%253A30&ticket=ST-114670-  
80 r00-cas'  
81 */  
82  
83 proc sort data=WORK.SORT out=work.sortDS  
84 (drop=DriveTrain Invoice MPG_City MPG_Highway Weight Wheelbase Length) equals;  
85 by MSRP;  
86 run;
```

NOTE: There were 20 observations read from the data set WORK.SORT.
NOTE: The data set WORK.SORTDS has 20 observations and 8 variables.

Messages: 2 User: u64183310

Obs	Make	Model	Type	Origin	MSRP	EngineSize	Cylinders	Horsepower
1	Acura	RSX Type S 2dr	Sedan	Asia	\$23,820	2.0	4	200
2	Audi	A4 1.8T 4dr	Sedan	Europe	\$25,940	1.8	4	170
3	Acura	TSX 4dr	Sedan	Asia	\$26,990	2.4	4	200
4	Audi	A4 3.0 4dr	Sedan	Europe	\$31,840	3.0	6	220
5	Acura	TL 4dr	Sedan	Asia	\$33,195	3.2	6	270
6	Audi	A4 3.0 Quattro 4dr manual	Sedan	Europe	\$33,430	3.0	6	220
7	Audi	A4 3.0 Quattro 4dr auto	Sedan	Europe	\$34,480	3.0	6	220
8	Audi	A4 1.8T convertible 2dr	Sedan	Europe	\$35,940	1.8	4	170
9	Audi	A6 3.0 4dr	Sedan	Europe	\$36,640	3.0	6	220
10	Acura	MDX	SUV	Asia	\$36,945	3.5	6	265
11	Audi	A6 3.0 Quattro 4dr	Sedan	Europe	\$39,640	3.0	6	220
12	Audi	A4 3.0 convertible 2dr	Sedan	Europe	\$42,490	3.0	6	220
13	Audi	A6 2.7 Turbo Quattro 4dr	Sedan	Europe	\$42,840	2.7	6	250
14	Acura	3.5 RL 4dr	Sedan	Asia	\$43,755	3.5	6	225
15	Audi	A4 3.0 Quattro convertible 2dr	Sedan	Europe	\$44,240	3.0	6	220
16	Acura	3.5 RL w/Navigation 4dr	Sedan	Asia	\$46,100	3.5	6	225
17	Audi	S4 Quattro 4dr	Sedan	Europe	\$48,040	4.2	8	340
18	Audi	A6 4.2 Quattro 4dr	Sedan	Europe	\$49,690	4.2	8	300
19	Audi	A8 L Quattro 4dr	Sedan	Europe	\$69,190	4.2	8	330
20	Acura	NSX coupe 2dr manual S	Sports	Asia	\$89,765	3.2	6	290

 **Data Smoothing:** A technique to reduce noise in data. Mainly used in datasets with date or time trend. Using point and click function to smooth data is easy: **Tasks and Utilities > Tasks > Data > Forecasting > Time Series Data Preparation**. Using smoothing in SAS makes it easier to analyse and visualize patterns and smooths out irregular fluctuations. In below mentioned example, using Time Series Data Preparation it gave us month of the event.

SAS® Studio

Server Files and Folders

Tasks and Utilities

- Tasks
 - Data
 - Graph
 - Map
 - Statistics
 - Linear Models
 - Survival Analysis
 - Multivariate Analysis
 - Cluster Analysis
 - Power and Sample Size
 - Statistical Process Control
 - Combinatorics and Probability
 - Data Mining
 - Econometrics
 - Forecasting
 - Time Series Data Preparation
 - Time Series Exploration
 - Modeling and Forecasting
 - Network Optimization
- Snippets
- Libraries
- File Shortcuts

*Time Series Data Preparation

CODE LOG RESULTS OUTPUT DATA

```
1 data smooth;
2 set sashelp.holiday;
3 run;
```

Line 3, Column 5 UTF-8
Messages: 6 User: u64183310

SAS® Studio

Settings Code/Results Split

DATA TRANSFORMATIONS OUTPUT INFORMATION

WORK.SMOOTH

Filter: (none)

ROLES

- *Time series variable
 - begin
 - end
 - rule
 - day

Treatment of missing values:
Missing value

ADDITIONAL ROLES

Time ID (1 item)
month

Properties

Interval: Day

Multi-line: 1

CODE LOG RESULTS OUTPUT DATA

```
10 ~ Generated on browser Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/104.0.0.0 Safari/537.36
11 * Generated on web client 'https://odamid-apsel-2.oda.sas.com/SASStudio/main?locale=en_GB&zone=...'
12
13 */
14
15 ods noproctitle;
16
17 proc sort data=WORK.SMOOTH out=work.preProcessedData;
18 by month;
19 run;
20
21 proc timedata data=work.preProcessedData seasonality=7 out=WORK._tsoutput;
22 id month interval=day setmissing=missing;
23 var begin / accumulate=none transform=none;
24 var end / accumulate=none transform=none;
25 var rule / accumulate=none transform=none;
26 var day / accumulate=none transform=none;
27 run;
28
29 data work.tsPrep(rename=());
30 set WORK._tsoutput;
31 run;
32
33 proc delete data=work.preProcessedData;
34 run;
35
36 proc delete data=WORK._tsoutput;
37 run;
```

Line 19, Column 5 UTF-8
Messages: 6 User: u64183310

SAS® Studio

Server Files and Folders

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- Libraries
- File Shortcuts

*Time Series Data Preparation

Settings Code/Results Split

DATA TRANSFORMATIONS C

WORK.SMOOTH

Filter: (none)

ROLES

- *Time series variable
 - begin
 - end
 - rule
 - day

Treatment of missing values:
Missing value

ADDITIONAL ROLES

Time ID (1 item)
month

Properties

Interval: Day

Multi-line: 1

CODE LOG RESULTS OUTPUT DATA

Errors, Warnings, Notes

Errors

Warnings

Notes (16)

```
98 data work.tsPrep(rename=());
99 set WORK._tsoutput;
100 run;
```

NOTE: There were 13 observations read from the data set WORK._TSI
NOTE: The data set WORK.TSPREP has 13 observations and 5 variables
NOTE: DATA statement used (Total process time):
real time 0.00 seconds
user cpu time 0.00 seconds
system cpu time 0.00 seconds
memory 943.15k
OS Memory 22700.00k
Timestamp 14/05/2025 02:35:51 AM
Step Count 74 Switch Count 2
Page Faults 0
Page Reclaims 125
Page Swaps 0
Voluntary Context Switches 10
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 264

Messages: 6 User: u64183310

Obs	month	begin	end	rule	day
1	01JAN1960	.	.	11	0
2	02JAN1960	1986	.	D	D
3	03JAN1960	1971	.	D	D
4	04JAN1960	.	.	.	.
5	05JAN1960	.	.	.	.
6	06JAN1960	1971	.	D	D
7	07JAN1960	.	.	1	3
8	08JAN1960	.	.	D	D
9	09JAN1960	.	.	.	.
10	10JAN1960	.	.	D	D
11	11JAN1960	1971	1978	D	D
12	12JAN1960	D	1971	D	D
13	13JAN1960	.	.	D	D

 **Data Standardization:** In SAS used to transform numeric variables into a standard format. It improves the comparability of variables. Using Standardize Data function as point and click function gives us standardized values of selected variables: **Tasks and Utilities > Tasks > Data > Standardize Data**. Various methods can be used under this function such as mean, standard deviation, range, sum, minkowski, etc.

In below mentioned example I have used standard deviation to get standardized values of CqRatio, Eqratio, NOxRatio.

SAS® Studio

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 - Recode Values
 - Recode Ranges
 - Combine Tables
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- Libraries
- File Shortcuts

*Program 1 *Standardize Data

CODE LOG RESULTS OUTPUT DATA

```
1 data stand;
2 set sashe1p.gas(obs=20);
3 run;
```

Line 2, Column 23 UTF-8

SAS® Studio

Server Files and Folders

Tasks and Utilities

- My Tasks
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 - Select Random Sample
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 - Rank Data
 - Transform Data
 - Standardize Data**
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 - Recode Ranges
 - Combine Tables
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- Libraries
- File Shortcuts

*Program 1 *Standardize Data

Settings Code/Results Split

DATA OPTIONS OUTPUT INFORMATION

DATA

WORK.STAND

Filter: (none)

ROLES

Variables to standardize:

- CpRatio
- EqRatio
- NOx

ADDITIONAL ROLES

CODE LOG RESULTS OUTPUT DATA

```
1 /*
2 *
3 * Task code generated by SAS Studio 3.8
4 *
5 * Generated on '14/05/2025 08:14'
6 * Generated by 'u64183310'
7 * Generated on server 'ODAWS02-APSE1-2.ODA.SAS.COM'
8 * Generated on SAS platform 'Linux X64 5.14.0-284.30.1.el9_
9 * Generated on SAS version '9.04.01M7P08062020'
10 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x6
11 * Generated on web client 'https://odamid-apse1-2.oda.sas.com/S
12 */
13
14
15 ods noproctitle;
16
17 proc stdize data=WORK.STAND method=std nomiss out=work.Stdize op
18 sprefix=Standardized_;
19 var CpRatio EqRatio NOx;
20 run;
```

Messages: 8 User: u64183310 UTF-8

SAS® Studio

Server Files and Folders

Tasks and Utilities

- My Tasks
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 - Characterize Data
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 - List Data
 - Transpose Data
 - Stack/Split Columns
 - Filter Data
 - Select Random Sample
 - Partition Data
 - Sort Data
 - Rank Data
 - Transform Data
 - Standardize Data**
 - Recode Values
 - Recode Ranges
 - Combine Tables
- Snippets
- Libraries
- File Shortcuts

*Program 1 *Standardize Data

Settings Code/Results Split

DATA OPTIONS OUTPUT INFORMATION

DATA

WORK.STAND

Filter: (none)

ROLES

Variables to standardize:

- CpRatio
- EqRatio
- NOx

ADDITIONAL ROLES

CODE LOG RESULTS OUTPUT DATA

LOG

Errors, Warnings, Notes

Errors

Warnings

Notes (4)

```
14 * Task code generated by SAS Studio 3.8
15 *
16 * Generated on '14/05/2025 08:14'
17 * Generated by 'u64183310'
18 * Generated on server 'ODAWS02-APSE1-2.ODA.SAS.COM'
19 * Generated on SAS platform 'Linux X64 5.14.0-284.30.1.el9_2.x86_64'
20 * Generated on SAS version '9.04.01M7P08062020'
21 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
22 Chrome/136.0.0.0 Safari/537.36'
23 * Generated on web client
24 'https://odamid-apse1-2.oda.sas.com/SASStudio/main?locale=en_GB&zone=GMT%252B05%253A3
25 rOO-cas'
26 *
27 */
28
29 ods noproctitle;
30
31 proc stdize data=WORK.STAND method=std nomiss out=work.Stdize oprefix
32 sprefix=Standardized_;
33 var CpRatio EqRatio NOx;
34 run;
```

NOTE: There were 20 observations read from the data set WORK.STAND.

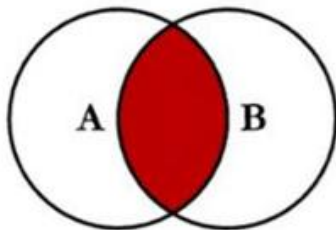
NOTE: The data set WORK.STDIZE has 20 observations and 7 variables.

Messages: 8 User: u64183310

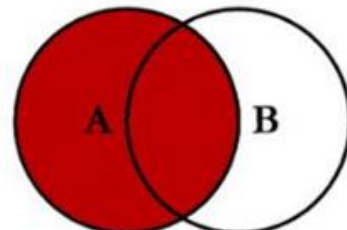
Obs	Fuel	CpRatio	EqRatio	NOx	Standardized_CpRatio	Standardized_EqRatio	Standardized_NOx
1	Ethanol	12.0	0.907	3.741	0.02995	-0.40988	1.58369
2	Ethanol	12.0	0.761	2.295	0.02995	-1.18503	0.29176
3	Ethanol	12.0	1.108	1.498	0.02995	0.65729	-0.42032
4	Ethanol	12.0	1.016	2.881	0.02995	0.16883	0.81532
5	Ethanol	12.0	1.189	0.760	0.02995	1.08734	-1.07969
6	Ethanol	9.0	1.001	3.120	-1.16810	0.08920	1.02886
7	Ethanol	9.0	1.231	0.638	-1.16810	1.31033	-1.18869
8	Ethanol	9.0	1.123	1.170	-1.16810	0.73693	-0.71338
9	Ethanol	12.0	1.042	2.358	0.02995	0.30688	0.34804
10	Ethanol	12.0	1.215	0.606	0.02995	1.22538	-1.21728
11	Ethanol	12.0	0.930	3.669	0.02995	-0.28776	1.51936
12	Ethanol	12.0	1.152	1.000	0.02995	0.89089	-0.86526
13	Ethanol	15.0	1.138	0.981	1.22800	0.81656	-0.88224
14	Ethanol	18.0	0.601	1.192	2.42604	-2.03451	-0.69372
15	Ethanol	7.5	0.696	0.926	-1.76712	-1.53013	-0.93138
16	Ethanol	12.0	0.686	1.590	0.02995	-1.58322	-0.33813
17	Ethanol	12.0	1.072	1.806	0.02995	0.46615	-0.14514
18	Ethanol	15.0	1.074	1.962	1.22800	0.47677	-0.00576
19	Ethanol	15.0	0.934	4.028	1.22800	-0.26652	1.84011
20	Ethanol	9.0	0.808	3.148	-1.16810	-0.93549	1.05387

Q2) Explain the illustrate merges on below images. Combine datasets using PROC SQL (optional)

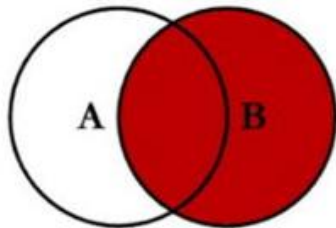
INNER MERGE / JOIN



LEFT OUTER MERGE / JOIN



RIGHT OUTER MERGE / JOIN



FULL OUTER MERGE / JOIN

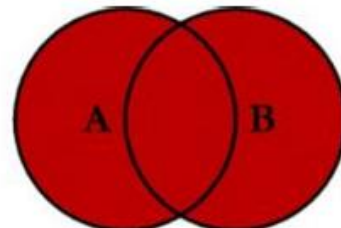
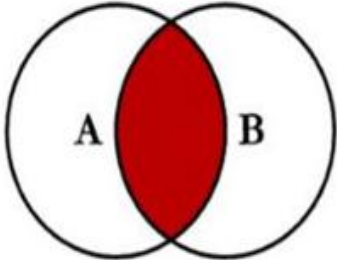
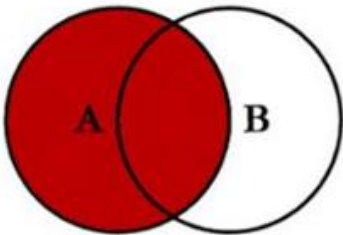
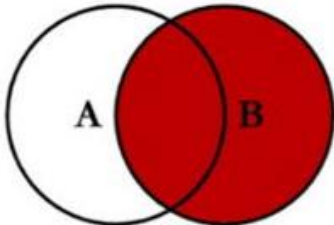


Illustration of Merges / Joins using Venn Diagrams

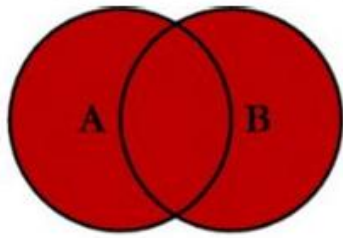
Ans2) Above mentioned diagrams are Venn diagrams of various joins or merges available in SAS studios. There are four different joins inner merge, left outer merge, right outer merge, full outer merge.

 **Inner Merge:** Also called as Inner Join, it combines rows from two or more datasets only where the key variable matches in both datasets. In SAS studio it can be accessed using 'PROC SQL' statement or through point and click function: **Tasks and Utilities > Tasks > Data > Combine Tables** or through this way SAS studio allows user to select two tables. Or "inner_join" under 'PROC SQL' allows user to combine two or more datasets.

 **Left Outer Merge:** Combines two datasets based on a specified join condition, including all rows from the left dataset and only matching rows from the right dataset. If there's no match, the result includes missing values from the right datasets. Same as inner merge point and click this merge can be accessed or "left_join" statement under 'PROC SQL' gives left outer merge.

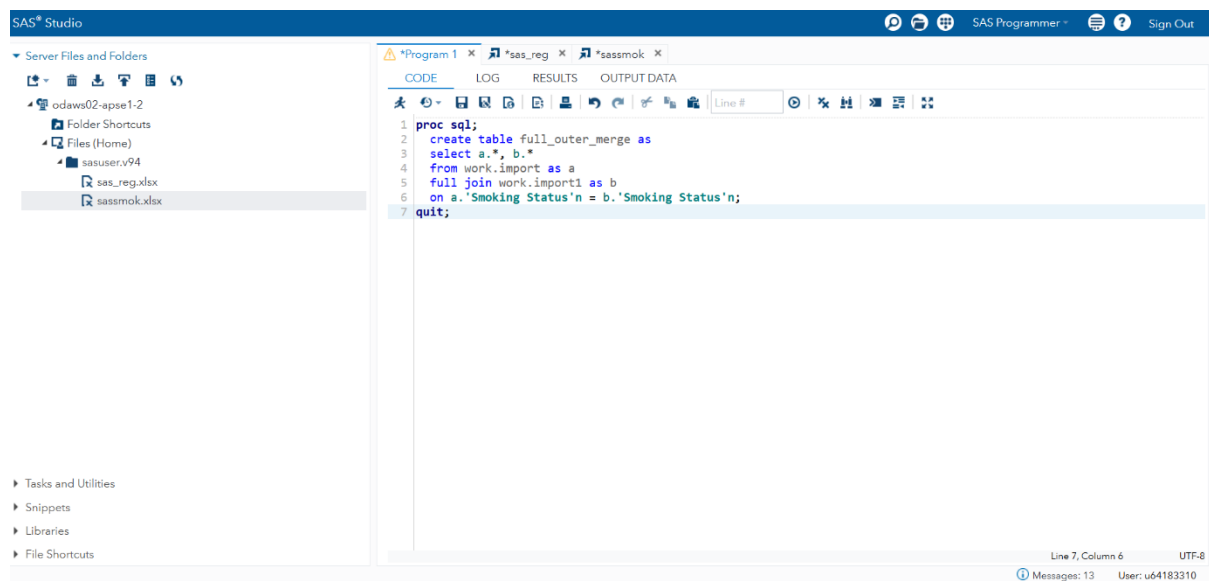
 **Right Outer Merge:** works just opposite to left outer merge it returns all rows from the right table, along with the matching rows from the left table and any unmatched rows from the right table. Using "right_join" statement under 'PROC SQL' gives this merge.

 **Full Outer Merge:** Full Outer Join combine all observations from two datasets. If a row in one dataset does not have a matching row in the other, the result will contain missing values from the non-matching side. Same as other joining point and click function is via **Tasks and Utilities > Tasks > Data > Combine Tables**. Or using “full_outer” under ‘PROC SQL’ statement, user can pragmatically perform full outer merge.



Below mentioned example shows combination of two separate datasets into one dataset, using “full Join” statement under ‘PROC SQL’

Method.



```
1 proc sql;
2   create table full_outer_merge as
3   select a.*, b.*
4   from work.import as a
5   full join work.import1 as b
6   on a.'Smoking Status' = b.'Smoking Status';
7 quit;
```

The screenshot shows the SAS Studio interface. On the left, the 'Server Files and Folders' pane lists files like 'sas_reg.xlsx' and 'sassmok.xlsx'. The main window displays a SAS code editor with a PROC SQL query that performs a full outer join between 'work.import' and 'work.import1' based on the 'Smoking Status' variable. The status bar at the bottom indicates 'Line 7, Column 6' and 'UTF-8' encoding.

The screenshot shows the SAS Studio interface. On the left is the 'Server Files and Folders' pane with a tree view showing folders like 'odaws02-apse1-2' and files like 'sas_reg.xlsx' and 'sassmok.xlsx'. The main area is divided into 'CODE', 'LOG', 'RESULTS', and 'OUTPUT DATA' tabs. The 'CODE' tab is active, showing a SAS program that creates a table 'full_outer_merge' using a full join of 'work.import' and 'work.import1' on 'Smoking Status'. The 'LOG' tab shows the execution output, including warnings about existing variables and a note about the table creation. The 'RESULTS' tab shows the 'PROCEDURE SQL used' statistics.

```

69 proc sql;
70   create table full_outer_merge as
71   select a.*, b.*
72   from work.import as a
73   full join work.import1 as b
74   on a.'Smoking Status' = b.'Smoking Status';
75 quit;

```

WARNING: Variable Sex already exists on file WORK.FULL_OUTER_MERGE.
 WARNING: Variable 'Smoking Status' already exists on file WORK.FULL_OUTER_MERGE.
 WARNING: Variable Height already exists on file WORK.FULL_OUTER_MERGE.
 WARNING: Variable Weight already exists on file WORK.FULL_OUTER_MERGE.
 NOTE: Table WORK.FULL_OUTER_MERGE created, with 15 rows and 15 columns.

NOTE: PROCEDURE SQL used (Total process time):
 real time 0.00 seconds
 user cpu time 0.00 seconds
 system cpu time 0.00 seconds
 memory 5919.53k
 OS Memory 28080.00k
 Timestamp 14/05/2025 02:56:35 AM
 Step Count 134 Switch Count 2
 Page Faults 0
 Page Reclaims 481
 Page Swaps 0
 Voluntary Context Switches 11

14/05/2025, 08:26

Results: WORK.FULL_OUTER_MERGE

Obs	Med Cnt ID	Age CHD Diag	Age At Start	Height	Weight	Diastolic	Systolic	MRW	Smoking Status	Age At Death	Cholesterol	Sex	Death Cause	N Obs	N
1	.	.	.	.	.	.	.	.	.	.	.	.	.	862	862
2	.	.	.	.	.	.	.	.	.	.	.	.	.	786	786
3	1	45	39	152	75	75	121	32.5	Heavy Smoker	51	190	Male	Chronic HD	.	.
4	.	.	.	.	.	.	.	.	.	.	.	.	.	1672	1672
5	.	.	.	.	.	.	.	.	.	.	.	.	.	1780	1780
6	2	36	32	165	68	72	119	25	Light Smoker	0	225	Female	None	.	.
7	.	.	.	.	.	.	.	.	.	.	.	.	.	321	321
8	.	.	.	.	.	.	.	.	.	.	.	.	.	236	236
9	3	41	36	160	73	80	120	28.5	Moderate Smoker	45	200	Female	Chronic HD	.	.
10	5	30	29	163	70	80	120	27	Non Smoker	0	221	Male	None	.	.
11	4	32	30	165	76	76	115	27.9	Non Smoker	0	230	Male	None	.	.
12	.	.	.	.	.	.	.	.	.	.	.	.	.	1682	1682
13	.	.	.	.	.	.	.	.	.	.	.	.	.	1983	1983
14	.	.	.	.	.	.	.	.	.	.	.	.	.	865	865
15	.	.	.	.	.	.	.	.	.	.	.	.	.	296	296

Q3) Create a Program on Binary Logistic Regression? With creation of example?

Ans3) Example on Binary Logistic Regression:

SAS® Studio

Server Files and Folders

Tasks and Utilities

- My Tasks
- Tasks
 - Data
 - Graph
 - Map
 - Statistics
 - Linear Models
 - One-Way ANOVA
 - Nonparametric One-Way ANOVA
 - N-Way ANOVA
 - Analysis of Covariance
 - Linear Regression
 - Binary Logistic Regression**
 - Predictive Regression Models
 - Generalized Linear Models
 - Mixed Models
 - Partial Least Squares Regression
 - Survival Analysis
 - Multivariate Analysis
- Snippets
- Libraries
- File Shortcuts

*sassmok *Binary Logistic Regression

Settings | Code/Results | Split

DATA MODEL SELECTION

Classification variables:

- Smoking Status

Parameterization of Effects

Coding:

GLM coding

Treatment of Missing Values

Continuous variables:

- Height
- Weight

CODE LOG RESULTS

```

1 /*
2 *
3 * Task code generated by SAS Studio 3.8
4 *
5 * Generated on '14/05/2025 09:03'
6 * Generated by 'u64183310'
7 * Generated on server 'ODAM502-APSE1-2.ODA.SAS.COM'
8 * Generated on SAS platform 'Linux X64 5.14.0-284.30.1.el9_
9 * Generated on SAS version '9.04.01M7P08062020'
10 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x6
11 * Generated on web client 'https://odamid-apse1-2.oda.sas.com/S
12 */
13
14 ods noproctitle;
15 ods graphics / imagemap=on;
16
17
18 proc logistic data=WORK.IMPORT2 plots=(effect);
19 class 'Smoking Status' n / param=glm;
20 model Sex(event='Female')=Height Weight 'Smoking Status' n /
21 selection=backward slstay=0.05 hierarchy=single details
22 run;
  
```

UTF-8

Messages: 15 User: u64183310

SAS® Studio

Server Files and Folders

Tasks and Utilities

- My Tasks
- Tasks
 - Data
 - Graph
 - Map
 - Statistics
 - Linear Models
 - One-Way ANOVA
 - Nonparametric One-Way ANOVA
 - N-Way ANOVA
 - Analysis of Covariance
 - Linear Regression
 - Binary Logistic Regression**
 - Predictive Regression Models
 - Generalized Linear Models
 - Mixed Models
 - Partial Least Squares Regression
 - Survival Analysis
 - Multivariate Analysis
- Snippets
- Libraries
- File Shortcuts

*sassmok *Binary Logistic Regression

Settings | Code/Results | Split

DATA MODEL SELECTION

MODEL EFFECTS

Select a model type:

- ☐ Main effects model
- ☐ Main effects and all n-way interactions
- ☒ Custom model

Model Effects

- Intercept
- Height
- Weight
- Smoking Status

Offset variable: (1 item)

Column

CODE LOG RESULTS

```

74 * Generated on '14/05/2025 09:03'
75 * Generated by 'u64183310'
76 * Generated on server 'ODAM502-APSE1-2.ODA.SAS.COM'
77 * Generated on SAS platform 'Linux X64 5.14.0-284
78 * Generated on SAS version '9.04.01M7P08062020'
79 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0
79 ! Chrome/136.0.0.0 Safari/537.36'
80 * Generated on web client
80 ! 'https://odamid-apse1-2.oda.sas.com/SASStudio/main?lo
80 ! r00-cas'
81 *
82 */
83
84 ods noproctitle;
85 ods graphics / imagemap=on;
86
87 proc logistic data=WORK.IMPORT2 plots=(effect);
88 class 'Smoking Status' n / param=glm;
89 model Sex(event='Female')=Height Weight 'Smoking Stati
90 selection=backward slstay=0.05 hierarchy=single detai
91 run;
  
```

NOTE: PROC LOGISTIC is modeling the probability that Sex='Female'

Messages: 15 User: u64183310

Model Information		
Data Set	WORK.IMPORT2	
Response Variable	Sex	Sex
Number of Response Levels	2	
Model	binary logit	
Optimization Technique	Fisher's scoring	

Number of Observations Read	10
Number of Observations Used	10

Response Profile		
Ordered Value	Sex	Total Frequency
1	Female	5
2	Male	5

Probability modeled is Sex='Female'.

Backward Elimination Procedure

Class Level Information							
Class	Value	Design Variables					
Smoking Status	Heavy (16-25)	1	0	0	0	0	0
	Heavy (16-25)	0	1	0	0	0	0
	Light (1-5)	0	0	1	0	0	0
	Moderate (6-15)	0	0	0	1	0	0
	Non-smoker	0	0	0	0	1	0
	Very Heavy (>25)	0	0	0	0	0	1

Step 0. The following effects were entered:

Intercept Height Weight Smoking Status

Model Convergence Status
Complete separation of data points detected.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	15.863	16.004
SC	16.166	18.424
-2 Log L	13.863	0.004

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	13.8594	7	0.0537
Score	9.2512	7	0.2351
Wald	0.1092	7	1.0000

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Height	1	0.0013	0.9710
Weight	1	0.0326	0.8567
Smoking Status	5	0.0177	1.0000

<https://odamid-apse1-2.oda.sas.com/SASStudio/sasexec/submissions/2b6aec9c-1d81-4681-88d0-7076db9cc36c/results>

14/05/2025, 09:03

Results: Binary Logistic Regression

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	166.1	1738.2	0.0091	0.9239
Height		1	-0.4902	13.4779	0.0013	0.9710
Weight		1	-1.4532	8.0474	0.0326	0.8567
Smoking Status	Heavy (16-25)	1	-12.0833	132.7	0.0083	0.9274
Smoking Status	Heavy (16-25)	1	3.8540	109.1	0.0012	0.9718
Smoking Status	Light (1-5)	1	0.2710	4935.3	0.0000	1.0000
Smoking Status	Moderate (6-15)	1	1.4792	161.7	0.0001	0.9927
Smoking Status	Non-smoker	1	6.5569	90.0272	0.0053	0.9419
Smoking Status	Very Heavy (>25)	0	0	.	.	.

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Height	0.613	<0.001	>999.999
Weight	0.234	<0.001	>999.999
Smoking Status Heavy (16-25 vs Very Heavy (>25))	<0.001	<0.001	>999.999
Smoking Status Heavy (16-25) vs Very Heavy (>25)	47.183	<0.001	>999.999
Smoking Status Light (1-5) vs Very Heavy (>25)	1.311	<0.001	>999.999
Smoking Status Moderate (6-15) vs Very Heavy (>25)	4.389	<0.001	>999.999
Smoking Status Non-smoker vs Very Heavy (>25)	704.051	<0.001	>999.999

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	100.0	Somers' D	1.000
Percent Discordant	0.0	Gamma	1.000
Percent Tied	0.0	Tau-a	0.556
Pairs	25	c	1.000

Analysis of Effects Eligible for Removal			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Height	1	0.0013	0.9710
Weight	1	0.0326	0.8567
Smoking Status	5	0.0177	1.0000

Step 1. Effect Smoking Status is removed:

Model Convergence Status
Complete separation of data points detected.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	15.863	6.016
SC	16.166	6.924
-2 Log L	13.863	0.016

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	13.8465	2	0.0010
Score	7.5469	2	0.0230
Wald	0.2579	2	0.8790

<https://odamid-apse1-2.oda.sas.com/SASStudio/sasexec/submissions/2b6aec9c-1d81-4681-88d0-7076db9cc36>

14/05/2025, 09:03

Results: Binary Logistic Regression

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Height	1	0.0684	0.7937
Weight	1	0.1640	0.6855

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	490.6	1401.3	0.1226	0.7262
Height	1	-2.3272	8.8979	0.0684	0.7937
Weight	1	-2.1763	5.3734	0.1640	0.6855

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Height	0.098	<0.001	>999.999
Weight	0.113	<0.001	>999.999

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	100.0	Somers' D	1.000
Percent Discordant	0.0	Gamma	1.000
Percent Tied	0.0	Tau-a	0.556
Pairs	25	c	1.000

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
0.0082	5	1.0000

Analysis of Effects Eligible for Removal			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Height	1	0.0684	0.7937
Weight	1	0.1640	0.6855

Step 2. Effect Height is removed:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	15.863	9.986
SC	16.166	10.591
-2 Log L	13.863	5.986

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	7.8769	1	0.0050
Score	5.8834	1	0.0153
Wald	2.6046	1	0.1066

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq

<https://odamid-apse1-2.oda.sas.com/SASStudio/sasexec/submissions/2b6aec9c-1d81-4681-88d0-7076db9cc36c/results>

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Weight	1	2.6046	0.1066

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	28.8095	17.6594	2.6615	0.1028
Weight	1	-0.4647	0.2879	2.6046	0.1066

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Weight	0.628	0.357	1.105

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	92.0	Somers' D	0.840
Percent Discordant	8.0	Gamma	0.840
Percent Tied	0.0	Tau-a	0.467
Pairs	25	c	0.920

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
5.7621	6	0.4504

Analysis of Effects Eligible for Removal			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Weight	1	2.6046	0.1066

Step 3. Effect Weight is removed:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

-2 Log L	=	13.863
----------	---	--------

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	0	0.6325	0.0000	1.0000

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
9.2512	7	0.2351

Note: All effects have been removed from the model.

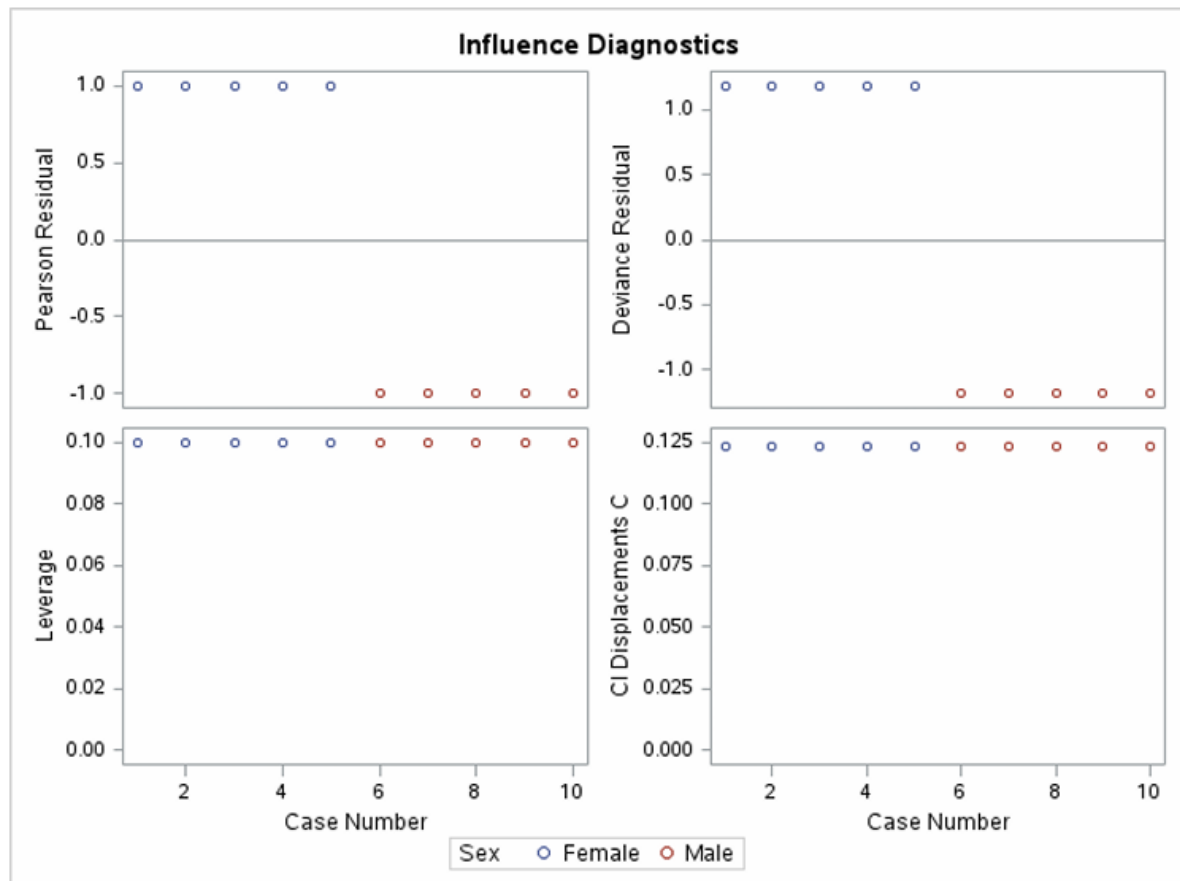
Summary of Backward Elimination						
Step	Effect Removed	DF	Number In	Wald Chi-Square	Pr > ChiSq	Variable Label
1	Smoking Status	5	2	0.0177	1.0000	Smoking Status
2	Height	1	1	0.0684	0.7937	Height

<https://odamid-apse1-2.oda.sas.com/SASStudio/sasexec/submissions/2b6aec9c-1d81-4681-88d0-7076db9cc36c/results>

14/05/2025, 09:03

Results: Binary Logistic Regression

Summary of Backward Elimination						
Step	Effect Removed	DF	Number In	Wald Chi-Square	Pr > ChiSq	Variable Label
3	Weight	1	0	2.6046	0.1066	Weight



Q4) Create a program on Simple Linear Regression with flow chart Presentation? With creation of example?

Ans4)

Load dataset into library



Tasks and Utilities



Tasks



Linear Models



Linear Regression



Select Data



Dependent Variable (weight)



Classification Variables (sex and Death Cause)



Continuous Variables (Diastolic and Systolic)



Model



Intercept Model (Diastolic, Systolic, Sex and Death Cause)



Run

SAS® Studio

Server Files and Folders

- odaws02-ape1-2
 - Folder Shortcuts
 - Files (Home)
 - sasuser.v94
 - sasbpnew.xlsx

Tasks and Utilities

- Snippets
- Libraries
- File Shortcuts

*Linear Regression

Settings | Code/Results | Split

DATA MODEL OPTIONS

WORKIMPORT3

Filter: (none)

ROLES

Dependent variable: (1 item)

- Weight

Classification variables:

- Sex
- Death Cause

Continuous variables:

- Diastolic
- Systolic

Parameterization of Effects

Treatment of Missing Values

CODE LOG RESULTS

Log Code

```
6 * Generated by 'u64183310'
7 * Generated on server 'ODAWS02-APSE1-2.ODA.SAS.COM'
8 * Generated on SAS platform 'Linux X64 5.14.0-284.30.1.el9'
9 * Generated on SAS version '9.04.01M7P08062020'
10 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x
11 * Generated on web client 'https://odamid-ape1-2.oda.sas.com/'
12
13 */
14
15 ods noproctitle;
16 ods graphics / imagemap=on;
17
18 proc glmselect data=WORK.IMPORT3 outdesign(addinputvars)=Work.r
19 class Sex 'Death Cause'n / param=glm;
20 model Weight=Diastolic Systolic Sex 'Death Cause'n / showp
21 selection=none;
22
23 run;
24
25 proc reg data=Work.reg_design alpha=0.05 plots(only)=(diagnosti
26 observedbypredicted);
27 where Sex is not missing and 'Death Cause'n is not missing;
28 ods select DiagnosticsPanel ResidualPlot ObservedByPredicted
29 model Weight=_GLSMOD /;
30 run;
31
32 quit;
33
34 proc delete data=Work.reg_design;
35 run;
```

Messages: 17 User: u64183310 UTF-8

SAS® Studio

Server Files and Folders

- odaws02-ape1-2
 - Folder Shortcuts
 - Files (Home)
 - sasuser.v94
 - sasbpnew.xlsx

Tasks and Utilities

- Snippets
- Libraries
- File Shortcuts

*Linear Regression

Settings | Code/Results | Split

DATA MODEL OPTIONS

MODEL EFFECTS

Model Effects

- Intercept
- Diastolic
- Systolic
- Sex
- Death Cause

CODE LOG RESULTS

Log Code

Errors, Warnings, Notes

- Errors
- Warnings
- Notes (7)

```
99 quit;
100
101 NOTE: PROCEDURE REG used (Total process time):
102 real time 0.75 seconds
103 user cpu time 0.18 seconds
104 system cpu time 0.05 seconds
105 memory 18056.06k
106 OS Memory 40680.00k
107 Timestamp 14/05/2025 03:51:46 AM
108 Step Count 167 Switch Count 22
109 Page Faults 0
110 Page Reclaims 14182
111 Page Swaps 0
112 Voluntary Context Switches 858
113 Involuntary Context Switches 8
114 Block Input Operations 0
115 Block Output Operations 960
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Results: Linear Regression

Data Set	WORK.IMPORT3
Dependent Variable	Weight
Selection Method	None

Number of Observations Read	5
Number of Observations Used	5

Class Level Information		
Class	Levels	Values
Sex	2	F M
Death Cause	2	None low BP

Dimensions	
Number of Effects	5
Number of Parameters	7

Least Squares Summary				
Step	Effect Entered	Number Effects In	Number Parm's In	SBC
0	Intercept	1	1	16.9357
1	Diastolic	2	2	18.4502
2	Systolic	3	3	19.9755
3	Sex	4	4	11.0279
4	Death Cause	5	5	-Infy*
* Optimal Value of Criterion				

Least Squares Model (No Selection)

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	107.20000	26.80000	.	.
Error	0	0	.		
Corrected Total	4	107.20000			

Root MSE	.
Dependent Mean	71.60000
R-Square	1.0000
Adj R-Sq	.
AIC	-Infty
AICC	-Infty
SBC	-Infty

Parameter Estimates					
Parameter	DF	Estimate	Standard Error	t Value	Pr > t
Intercept	1	682.000000	.	.	.
Diastolic	1	-1.000000	.	.	.
Systolic	1	-4.500000	.	.	.
Sex F	1	-5.500000	.	.	.
Sex M	0	0	.	.	.
Death Cause None	1	12.000000	.	.	.

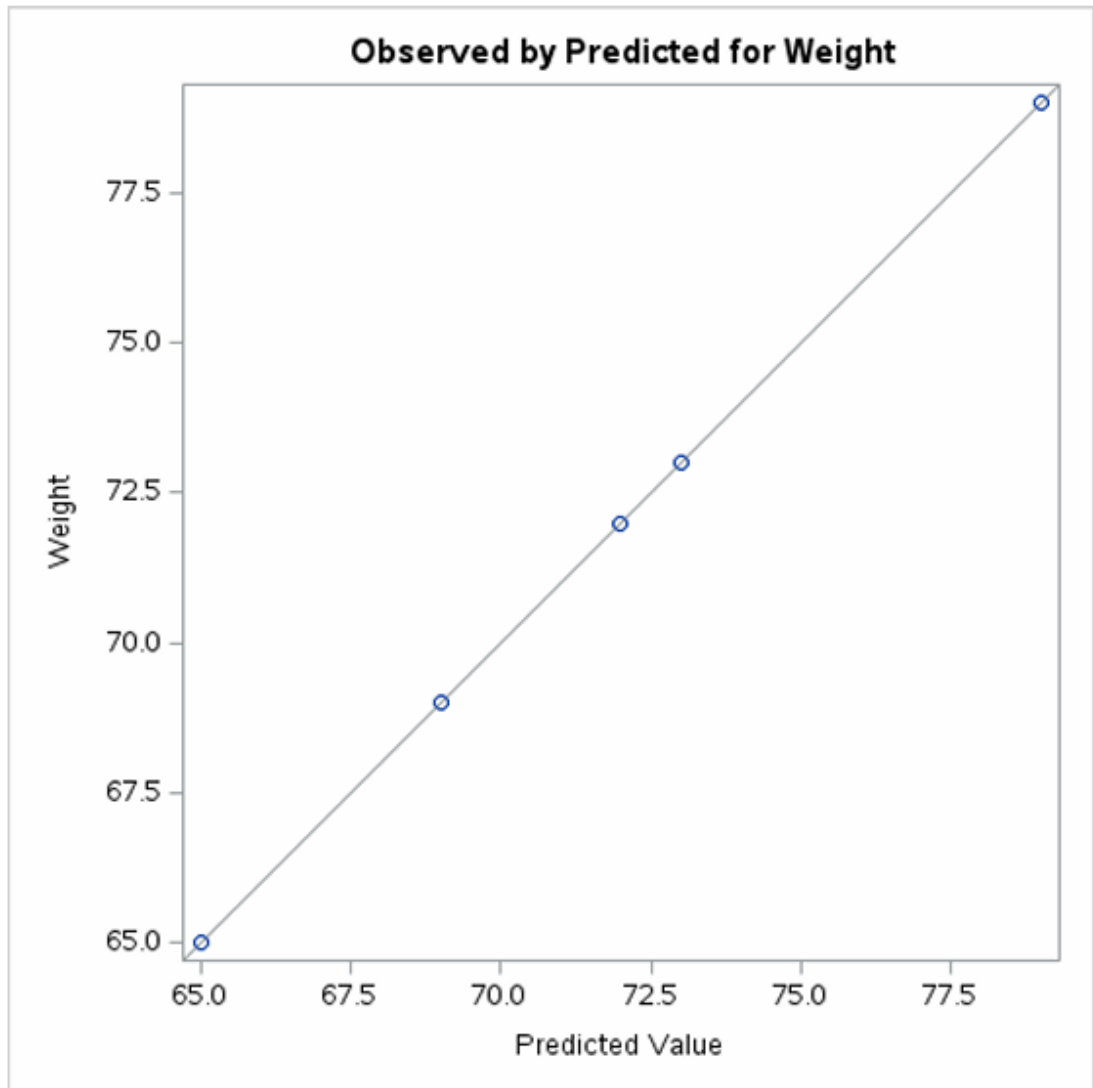
<https://odamid-apse1-2.oda.sas.com/SASStudio/sasexec/submissions/74142818-c817-4582-8c50-95778641a5e3/results>

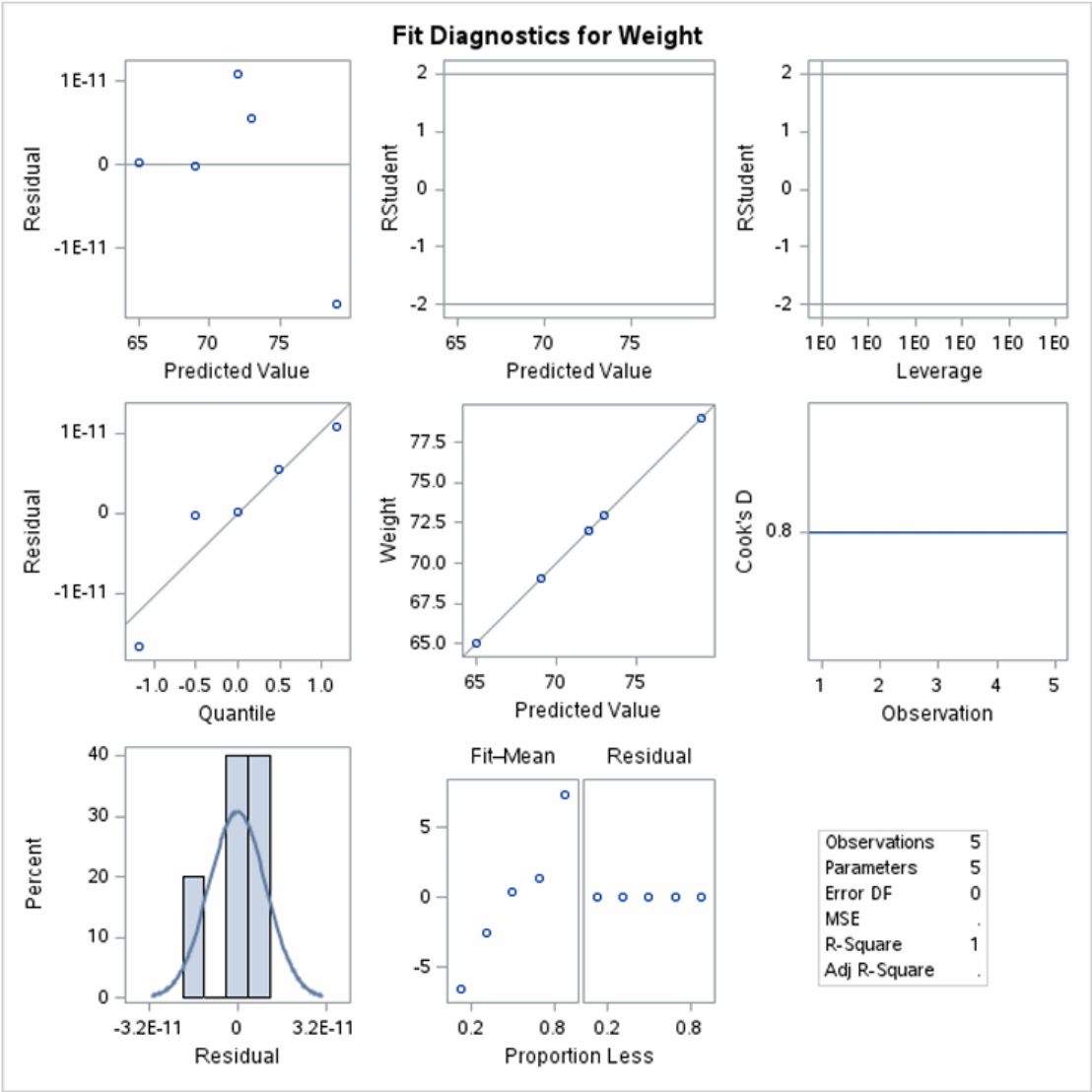
14/05/2025, 09:23

Results: Linear Regression

Parameter Estimates					
Parameter	DF	Estimate	Standard Error	t Value	Pr > t
Death Cause low BP	0	0	.	.	.

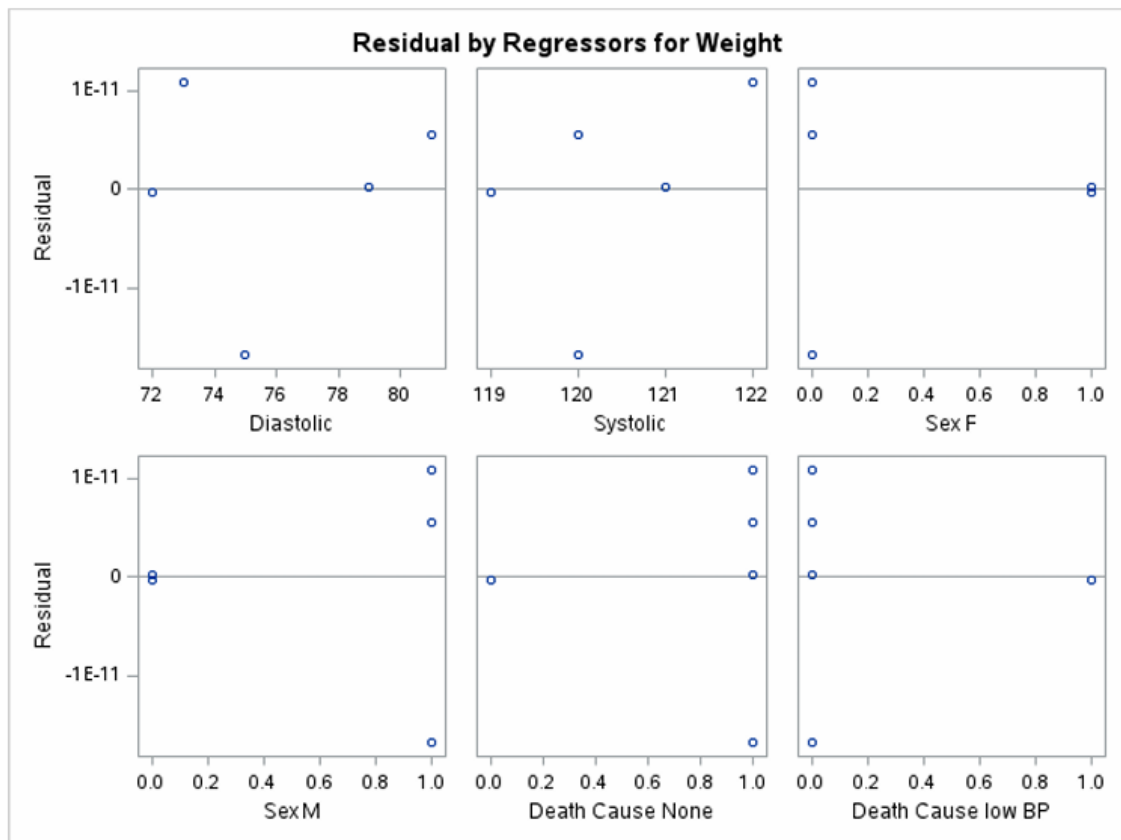
Model: MODEL1
Dependent Variable: Weight Weight





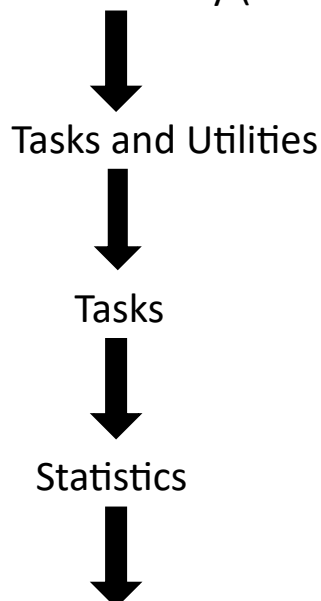
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Results: Linear Regression



Q5) Create a program on Data Exploration with Flow Chart Presentation? With creation of an example?

Ans5) Load dataset into library (using SASHELP dataset)



Data Exploration



Select Data



Continuous Variable (length 1,2,3)



Classification Variables (species and weight)



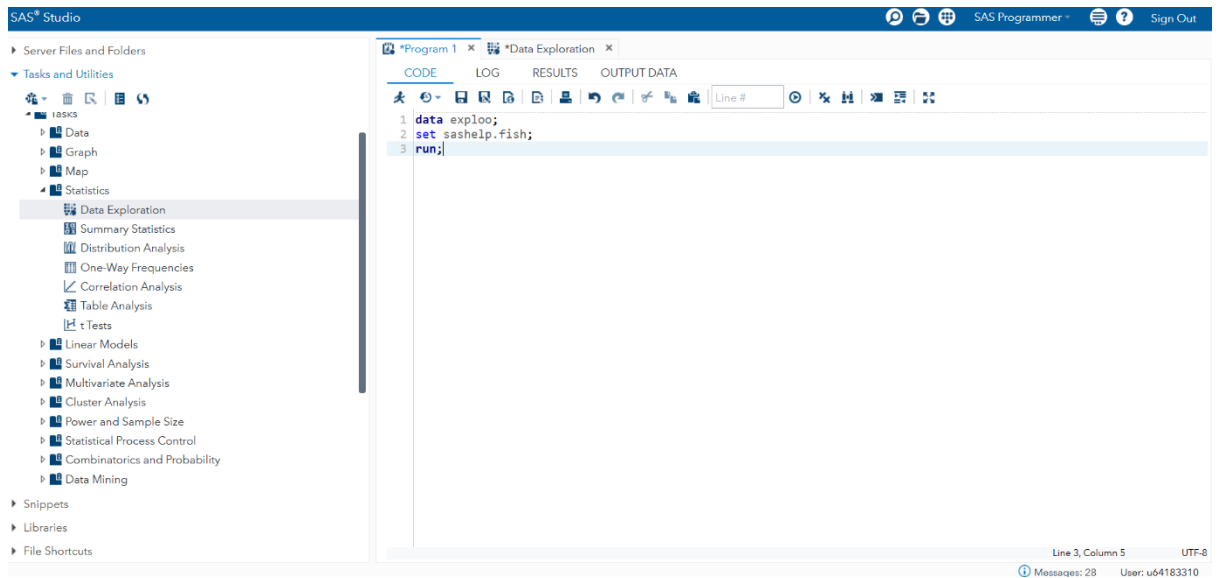
Add Scatter Plot Matrix



Add Regression Scatter Plot Matrix



Run



SAS® Studio

Server Files and Folders

Tasks and Utilities

- Tasks
 - Data
 - Graph
 - Map
 - Statistics
 - Data Exploration
 - Summary Statistics
 - Distribution Analysis
 - One-Way Frequencies
 - Correlation Analysis
 - Table Analysis
 - t Tests
 - Linear Models
 - Survival Analysis
 - Multivariate Analysis
 - Cluster Analysis
 - Power and Sample Size
 - Statistical Process Control
 - Combinatorics and Probability
 - Data Mining
- Snippets
- Libraries
- File Shortcuts

Program 1 *Data Exploration *

Settings | Code/Results | Split

DATA PLOTS INFORMATION

DATA

WORK.EXPLOO

ROLES

Continuous variables: (6 items)

- Length1
- Length2
- Length3

Classification variables: (2 items)

- Species
- Weight

CODE LOG RESULTS

```
1 * generated on SAS platform Linux L4N x86_64 5.14.0-284.el9
2 * Generated on SAS version '9.04.01M7P08062020'
3 * Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x
4 * Generated on web client 'https://odamid-apsel-2.oda.sas.com/
5 */
6
7 options validvarname=any;
8 ods noproctitle;
9 ods graphics / imagemap=on;
10
11 /* Scatter plot matrix macro */
12 %macro scatterPlotMatrix(xVars=, title=, groupVar=);
13
14 proc sgscatter data=WORK.EXPLOO;
15   matrix &xVars / %if(&groupVar ne %str()) %then
16     %do;
17       group=&groupVar legend=(sortorder=ascending) %e
18       diagonale(histogram kernel normal) ellipse=(type=predic
19       title &title;
20 run;
21
22 title;
23 %mend scatterPlotMatrix;
24
25 %scatterPlotMatrix(xVars=Length1 Length2 Length3,
26   title="Scatter plot matrix grouped by Species", groupVar=Sp
27 %scatterPlotMatrix(xVars=Length1 Length2 Length3,
28   title="Scatter plot matrix grouped by Weight", groupVar=Wei
29
30 UTF-8
```

Messages: 28 User: u64183310

SAS® Studio

Program 1 *Data Exploration *

Settings | Code/Results | Split

DATA PLOTS INFORMATION

SCATTER PLOT MATRIX

- ☒ Scatter plot matrix
- ☒ Add histograms
- ☒ Add normal density curve
- ☒ Add kernel density estimate
- ☒ Add prediction ellipses
- Probability level: 0.95

PAIRWISE SCATTER PLOTS

- ☐ Pairwise scatter plots

REGRESSION SCATTER PLOTS

- ☒ Regression scatter plots
- Select response variables

 - Length1
 - Length2
 - Length3

MOSAIC PLOT

CODE LOG RESULTS

Errors, Warnings, Notes

Errors

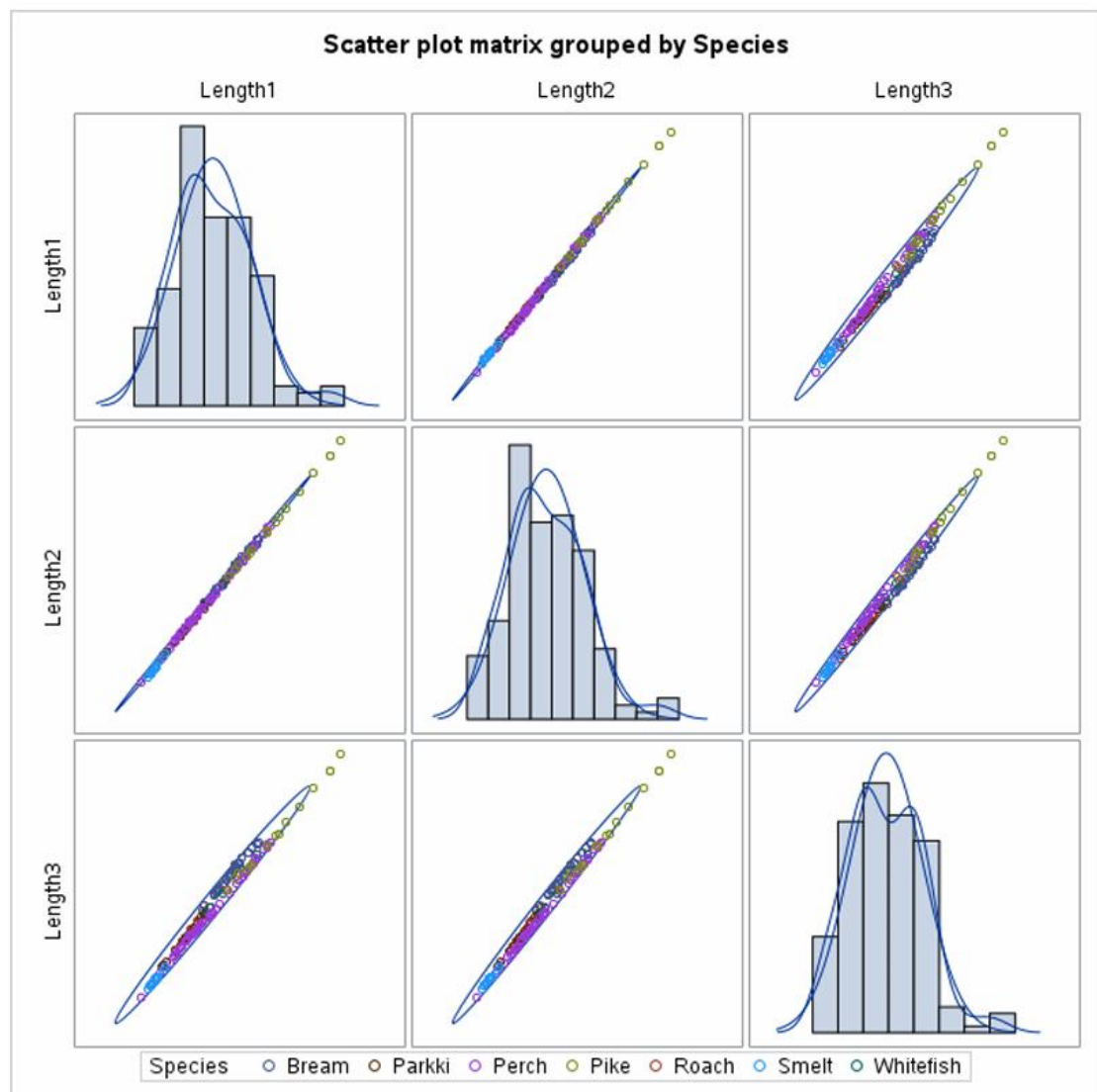
Warnings

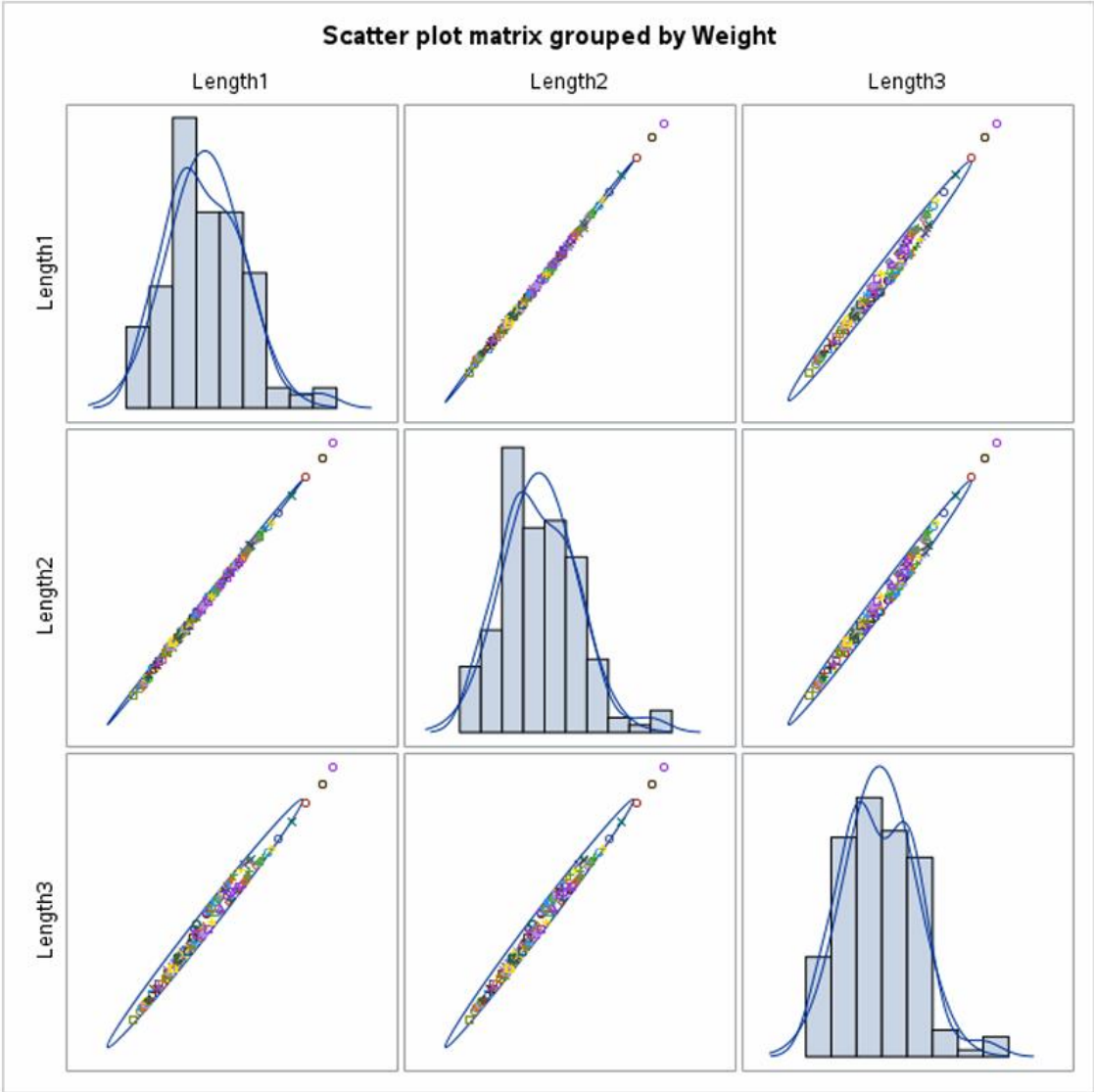
Notes (6)

```
97
98 title;
99 %mend scatterPlotMatrix;
100
101 %scatterPlotMatrix(xVars=Length1 Length2 Length3,
102   title="Scatter plot matrix grouped by Species", groupVar=Species);
NOTE: PROCEDURE SGSCATTER used (Total process time):
real time      0.36 seconds
user cpu time   0.17 seconds
system cpu time 0.01 seconds
memory         10534.03k
OS Memory      34996.00k
Timestamp      14/05/2025 04:30:53 AM
Step Count     273  Switch Count  2
Page Faults    0
Page Reclaims  2176
Page Swaps     0
Voluntary Context Switches 1286
Involuntary Context Switches 5
Block Input Operations 0
Block Output Operations 1664

NOTE: There were 159 observations read from the data set WORK.EXPLOO.
```

Messages: 28 User: u64183310





-----THE END-----